

Answer Key — Biology Grade 11

Sub-Strand 1.2: Ecology • Lesson 7: Factors Affecting Population Size: Why Are the Fish Disappearing?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	A	C	A	A	B	D

USE AFTER: PREDICT PHASE — Predict: pair-share and sort ideas into 'living things' and 'non-living things'

Q1. Learners sorted factors that could reduce fish numbers in a Kenyan river into 'living' and 'non-living'. Which choice correctly gives one biotic factor and one abiotic factor?

- A. Biotic: water temperature; abiotic: disease and parasites
- B. Biotic: drought lowering water level; abiotic: algae cover
- C. Biotic: dissolved oxygen; abiotic: competition among tilapia
- D. Biotic: Nile perch preying on small fish; abiotic: low dissolved oxygen in the water**

Correct answer: D. Predation by Nile perch involves living organisms, so it is biotic. Dissolved oxygen is a non-living part of the environment, so it is abiotic. The other choices swap the two categories.

USE AFTER: OBSERVE PHASE — Part A: Khan Academy reading on population growth and limiting factors (meaning of carrying capacity)

Q2. After reading about population growth and limiting factors, which statement best describes the carrying capacity of a river for tilapia?

- A. The maximum number of tilapia that the river's resources can support over time**
- B. The number of tilapia that fishermen are allowed to catch each season
- C. The number of tilapia born in the river each year
- D. The number of different fish species living in the river

Correct answer: A. Carrying capacity is the maximum population size an environment can support, given food, space, oxygen and other resources. It is not a birth rate, a fishing quota or a count of species.

USE AFTER: OBSERVE PHASE — Part B: PhET simulation, trial 2 (adding predators) with one variable changed at a time

Q3. In a simulation, a group first runs a baseline with fish and food only. In trial 2 they add predators and change nothing else. The fish population ends lower than in the baseline. Why can they conclude that predators caused the drop?

- A. Because the graph in trial 2 has more lines than the graph in the baseline
- B. Because predators are always the most important factor in any ecosystem
- C. Because only one variable was changed, so the predators are the only difference between the trials**
- D. Because the fish population was lower at the start of trial 2

Correct answer: C. When only one variable is changed per trial, any difference in the results can be linked to that variable. This is why learners were told to change only ONE thing at a time.

USE AFTER: OBSERVE PHASE — Part B: PhET simulation, trial 3 (reducing food)

Q4. In the PhET simulation, learners keep the fish but reduce the amount of food available. What do they most likely observe on the fish population graph?

- A. The fish population stops growing and falls to a lower level, because food is a limiting factor**
- B. The fish population rises quickly, then the food increases
- C. The fish population keeps rising because there are no predators
- D. The fish population stays exactly the same as the baseline

Correct answer: A. Food availability limits how many fish can survive. With less food the environment supports fewer fish, so the population is lower than in the baseline.

USE AFTER: EXPLAIN PHASE — Discussion Diamond: ranking the nine factor cards and claim-evidence-reasoning response 'Why are fish disappearing while algae increase?'

Q5. A river receives fertiliser and sewage from farms and homes upstream. Fish numbers fall while algae and weeds increase and the water smells bad. Which explanation best links these observations?

- A. Excess nutrients cause an algae boom. When the algae die and decompose, dissolved oxygen falls and fish cannot survive.**
- B. Algae cover lets more light reach the bottom of the river, which makes the fish leave.
- C. Algae are poisonous, so they kill the fish directly as soon as they appear.
- D. Extra nutrients give the fish more food, so the fish grow old and die in large numbers.

Correct answer: A. Nutrients feed algae, which then die and decompose. Decomposition uses up dissolved oxygen, so fish suffocate. Algae cover also reduces light, so it does not increase it.

USE AFTER: EXPLAIN PHASE — Discussion Diamond: oral question 'Why can a population not grow forever?'

Q6. Tilapia in a dam breed every few months, yet their population stops increasing at about 5 000 fish and stays near that number. What is the best explanation?

- A. The tilapia choose to stop breeding when the dam looks full
- B. Limiting factors such as food, oxygen and space prevent the population from growing beyond what the dam can support**
- C. Population size can only grow until it reaches 5 000 in any water body
- D. Predators are the only factor that can control a population

Correct answer: B. Limiting factors (food, dissolved oxygen, space, predation, disease) set the carrying capacity, so no population can grow forever. More than one factor usually acts together, so predators are not the only control.

USE AFTER: MODEL BUILDING PHASE — Revising the ecosystem model of the river: adding human activity (fishing, fertiliser from farms, untreated sewage)

Q7. Learners are adding human activities to their river ecosystem model and linking them to nutrient input. Which human activity should they connect to extra nutrients in the water?

- A. Nile perch preying on small fish
- B. Drought lowering the water level
- C. Tilapia competing for food
- D. Fertiliser from farms and untreated sewage washing into the river**

Correct answer: D. Fertiliser runoff and untreated sewage are human activities that add nitrogen and other nutrients to water. Predation, drought and competition are natural factors and do not add nutrients.